

Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2026

Course: B.Tech
Code: 100218

Subject: Python Programming

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option for the following questions (Answer any seven):

[2 x 7 = 14]

- (a) Python uses _____ to define blocks of code
 - (i) Braces
 - (ii) Indentation
 - (iii) Parentheses
 - (iv) Semicolon
- (b) The operator used for exponentiation is
 - (i) ^
 - (ii) //
 - (iii) **
 - (iv) %
- (c) The default step value in slicing is
 - (i) 0
 - (ii) 1
 - (iii) -1
 - (iv) 2
- (d) Strings in Python are
 - (i) Mutable
 - (ii) Dynamic Arrays
 - (iii) Immutable
 - (iv) Numeric Objects
- (e) Negative indexing begins from
 - (i) First element
 - (ii) Last element
 - (iii) Middle element
 - (iv) Zero index
- (f) Dictionaries store data in the form of
 - (i) Indexes
 - (ii) Key-value pairs
 - (iii) Lists
 - (iv) Tuples
- (g) The function used to combine two sequences into pairs is
 - (i) join()
 - (ii) pair()
 - (iii) zip()
 - (iv) merge()
- (h) Pickle module is mainly used for
 - (i) Image processing++
 - (ii) Networking
 - (iii) Object serialization
 - (iv) Graphics
- (i) CSV stands for
 - (i) Common Separated Values
 - (ii) Character Separated Values
 - (iii) Comma Separated Values
 - (iv) Column Separated Variables
- (j) Python is known as a _____ typed language.
 - (i) Weakly
 - (ii) Statically
 - (iii) Dynamically
 - (iv) Manually

- Q.2 (a) Explain the different features of Python Programming language. What are the different arithmetic operators in Python? Explain their precedence and associativity. [7]
- (b) What are the basic data types in Python? How does Python decide the type of any variable? How a float is converted to an integer? Explain. [7]

- Q.3** (a) Evaluate the importance of user-defined functions in Python. Compare them with built-in functions and discuss situations where each should be preferred. [7]
- (b) Assess the significance of string slicing, indexing and formatting in Python. Demonstrate the output of the following statements and justify your answers. [7]
- ```
s = "Programming"
print(s[2:8])
print(s[-5:])
print(s[::-1])
```
- Q.4** (a) Differentiate between Lists, Tuples and Sets on the basis of syntax, mutability, indexing, performance and applications. Support your comparison with suitable examples. [7]
- (b) Construct a Python program that creates a dictionary containing employee ID, employee name and salary. Display all records and update the salary of a given employee using suitable dictionary methods. [7]
- Q.5** (a) Write a python program to find Greatest Common Divisor of two numbers. [7]
- (b) Illustrate with suitable examples the following list methods in Python. [7]
- (i) insert (ii) append (iii) pop (iv) extend
- Q.6** (a) Write the differences between the if, if-else and if-elif-else statements in Python. Provide code examples to demonstrate the usage and scenarios where each statement is preferred. [7]
- (b) Explain the utility of break and continue statement with an example. [7]
- Q.7** (a) What is dictionary and how it is created in Python? What is meant by key-value pairs in a dictionary? How does del operation work on dictionaries? Give an example. [7]
- (b) Write a python program to read two numbers. Then find out whether the first number is a multiple of the second number. [7]
- Q.8** (a) Design a Python program to create a text file, write the details of ten students into it, and subsequently read the stored data. Discuss the sequence of file operations involved in the program. [7]
- (b) Justify the significance of the os and os.path modules in Python. Demonstrate the use of suitable functions to create directories, determine file paths and verify file existence. [7]
- Q.9** Write Short notes on **any two** of the following: [2 x 7 = 14]
- Dynamic and Strongly Typed Language
  - Type Conversion
  - Scope and Lifetime of Variables
  - CSV File Handling